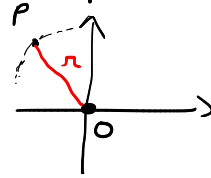


EXERCÍCIO 35, S. 1.2

DETERMINAR Eq. da circunferência (C)

Centro na origem: $(0,0)$

$P = (-3, 5) \in C$.



$$d(P, O) = r$$

Eq. geral da circunferência de centro (h, k) e raio r :

$$(x-h)^2 + (y-k)^2 = r^2$$

Logo, eq. C:

$$(x-0)^2 + (y-0)^2 = r^2$$

$$x^2 + y^2 = r^2$$

Onde $r = d(P, O)$ $P = (-3, 5)$ $O = (0, 0)$

$$= \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$$= \sqrt{(-3)^2 + (5)^2}$$

$$= \sqrt{9 + 25} = \sqrt{34}$$

$$x^2 + y^2 = (\sqrt{34})^2$$

$$x^2 + y^2 = 34$$

LEMBRE-SE:

$$(x+a)^2 = x^2 + 2ax + a^2$$

USAMOS PARA
"COMPLETAR
O QUADRADO"

$$(x \pm a)^2 = x^2 \pm 2ax + a^2$$

42)

$$\underline{x^2} + \underline{y^2} - \underline{10x} + \underline{2y} + 22 = 0$$

$$(x^2 - 10x) + (y^2 + 2y) + 22 = 0$$

$$\underbrace{(x^2 - 10x + 5^2)}_{(x-5)^2} + \underbrace{(y^2 + 2y + 1^2)}_{(y+1)^2} + 22 = 0 + 5^2 + 1^2$$

$$(x-5)^2 + (y+1)^2 + 22 = 26 \Rightarrow \boxed{(x-5)^2 + (y+1)^2 = 2^2}$$

-h = -5 $\Rightarrow$ h = 5
CENTRO: -k = +1 $\Rightarrow$ k = -1

$$\begin{cases} (5, -1) \\ \text{Raio: } 2 \end{cases}$$

43)

$$\underline{x^2} + \underline{y^2} + \underline{6x} = 0 \Rightarrow (x^2 + 6x) + y^2 = 0$$

$$\underbrace{(x^2 + 6x + 3^2)}_{(x+3)^2} + y^2 = 0 + 3^2$$

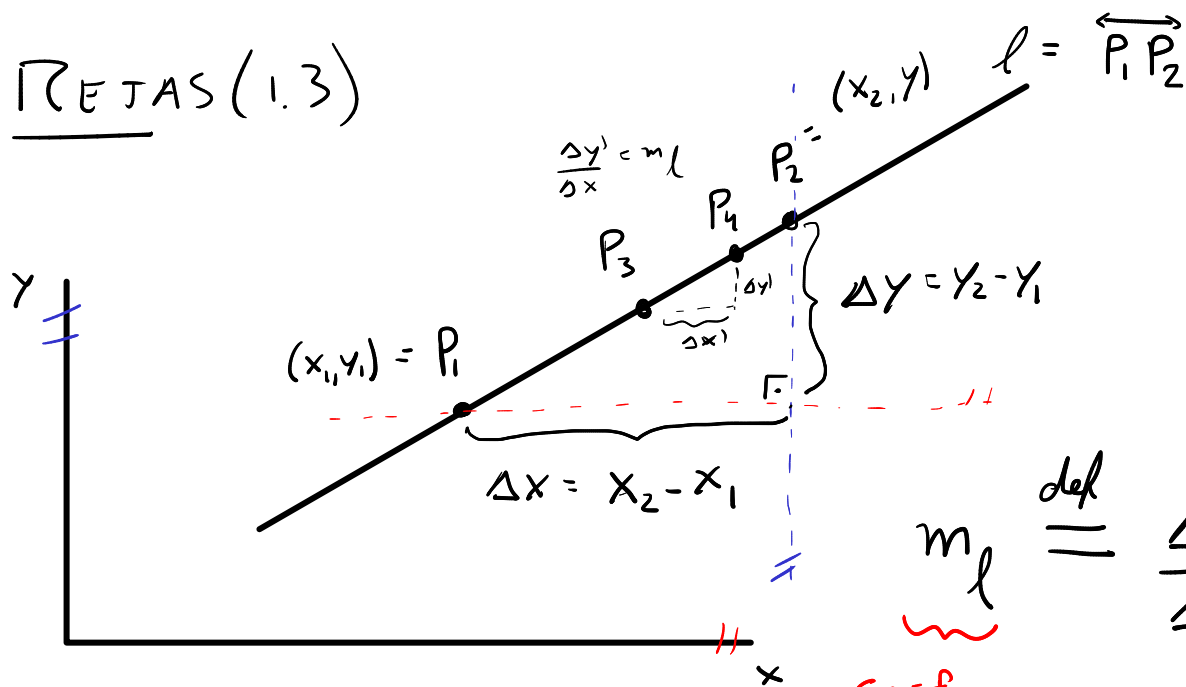
$$(x+3)^2 + (y-0)^2 = 9$$

Eq: $(x+3)^2 + (y-0)^2 = 3^2$

CENTRO: (-3, 0)

Raio: 3

RETAS (1.3)



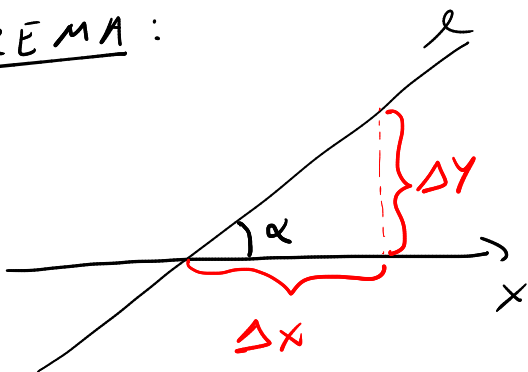
$$m_l \stackrel{\text{def}}{=} \frac{\Delta Y}{\Delta X} = \frac{Y_2 - Y_1}{X_2 - X_1}$$

coef. ANGULAR DA RETA $\sim$ TAXA DE VARIAÇÃO MÉDIA

CARACTERIZAÇÃO DA RETA $\left\{ \begin{array}{l} \text{É CONSTANTE} \\ \text{NA RETA} \end{array} \right.$

o coef. angular NÃO depende dos pontos da reta.

TEOREMA:



$$m_l = \text{Tg } \alpha$$

$$\text{Tg } \alpha \stackrel{\text{def}}{=} \frac{\Delta y}{\Delta x} \stackrel{\text{def}}{=} m_l$$

COROLÁRIO:

$$0 < \alpha < 90^\circ \Leftrightarrow \text{Tg } \alpha > 0 \Leftrightarrow \underline{m_l > 0}$$

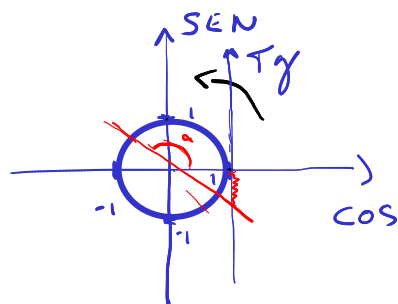
(RETA CRESCENTE)

$$90^\circ < \alpha < 180^\circ \Leftrightarrow \text{Tg } \alpha < 0$$

(RETA
DECRESCENTE)

$$\sim m_l < 0$$

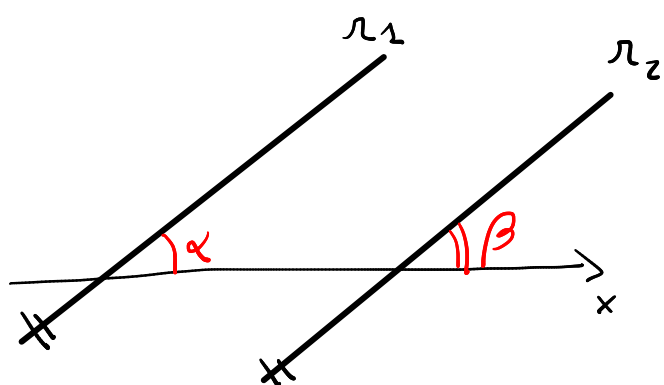
Obs: CICLO
TRIGONOMÉTRICO
CIRCUNFERÊNCIA NA ORIGEM
e RAIO 2



RETAS

PARALELAS

$$r_1 \parallel r_2 \Leftrightarrow \alpha = \beta$$

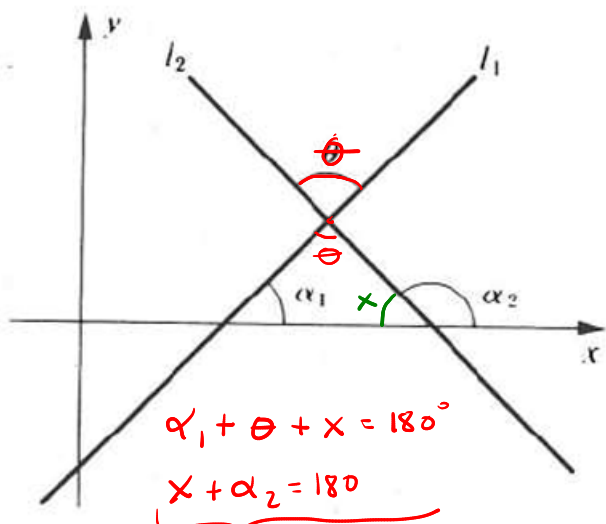


$$\left. \begin{array}{l} m_{r_1} = \text{Tg } \alpha \\ m_{r_2} = \text{Tg } \beta \end{array} \right\} m_{r_1} = m_{r_2}$$

Outro jeito, $r_1 \parallel r_2 \Leftrightarrow m_{r_1} = m_{r_2}$

TEOREMA: $l_1 \perp l_2 \Leftrightarrow m_{l_1} \cdot m_{l_2} = -1$

$\alpha_2 > \alpha_1$ (ASSUMINDO SEM PERDA DE GENERALIDADE)



$$\alpha_1 + \theta + x = 180^\circ$$

$$x + \alpha_2 = 180^\circ$$

$$\alpha_1 + \theta + x = x + \alpha_2 \Rightarrow \boxed{\theta = \alpha_2 - \alpha_1} \otimes$$

LEMBRETE:

$$\text{sen}(a \pm b) = \text{sen } a \cos b \pm \text{sen } b \cos a$$

$$\cos(a \pm b) = \cos a \cos b \mp \text{sen } a \text{sen } b$$

$$\text{Tg}(a-b) = \frac{\text{sen}(a-b)}{\cos(a-b)}$$

$$= \frac{\text{sen } a \cos b - \text{sen } b \cos a}{\cos a \cos b + \text{sen } a \text{sen } b} \div \cos a$$

$$= \frac{\cancel{\sin a} \cos b - \cancel{\sin b} \cos a}{\cancel{\cos a} \cos b + \cancel{\sin a} \cancel{\sin b}} = \frac{\cancel{\sin a} \cdot \cos b - \cancel{\sin b} \cdot \cos a}{\cos b + \cancel{\sin a} \cdot \cancel{\sin b}} \div \cos b$$

$$= \frac{\cancel{\sin a} \cos b - \cancel{\sin b}}{\cos b} \div \frac{\cos b + \cancel{\sin a} \cdot \cancel{\sin b}}{\cos b}$$

$$= \boxed{\frac{\cancel{\sin a} - \cancel{\sin b}}{1 + \cancel{\sin a} \cdot \cancel{\sin b}} = \text{Tg}(a-b)} \quad \textcircled{*}$$

USAMOS $\textcircled{*}$ em $\theta = \alpha_2 - \alpha_1$

$$\text{Tg } \theta = \text{Tg}(\alpha_2 - \alpha_1) \stackrel{(*)}{=} \frac{\text{Tg } \alpha_2 - \text{Tg } \alpha_1}{1 + \text{Tg } \alpha_2 \cdot \text{Tg } \alpha_1}$$

$\theta = 90^\circ \Rightarrow \nexists \text{Tg } 90^\circ$, mas por $\textcircled{*}$: $\text{Tg } \theta = \frac{\text{Tg } \alpha_2 - \text{Tg } \alpha_1}{1 + \text{Tg } \alpha_2 \cdot \text{Tg } \alpha_1}$

NÃO EXISTE

$\underbrace{\text{Tg } \alpha_2}_{m_{l_2}} \cdot \underbrace{\text{Tg } \alpha_1}_{m_{l_1}} = -1 \quad \underbrace{1 + \text{Tg } \alpha_2 \cdot \text{Tg } \alpha_1}_{=0}$

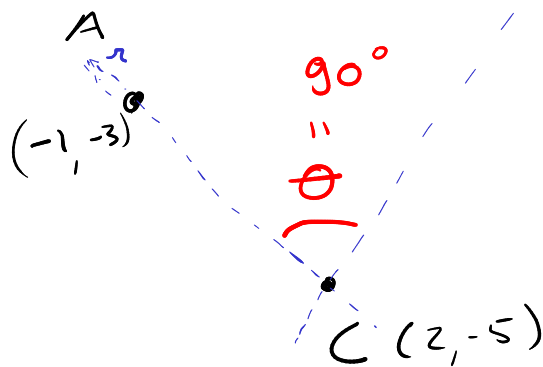
$$m_{l_1} \cdot m_{l_2} = -1$$

Logo, $l_1 \perp l_2 \Leftrightarrow m_{l_1} \cdot m_{l_2} = -1$

Exemplo 2

$$r = \overline{AC}$$
$$s = \overline{CB}$$

$$m_r = \frac{\Delta y}{\Delta x}$$
$$= \frac{-5 - (-3)}{2 - (-1)}$$
$$= \frac{-2}{3}$$



$$m_s = \frac{\Delta y}{\Delta x}$$
$$= \frac{1 - (-5)}{6 - 2}$$
$$= \frac{6}{4} = \frac{3}{2}$$

$$m_{\overline{AC}} = -\frac{2}{3} \quad \text{e} \quad m_{\overline{CB}} = \frac{3}{2}$$

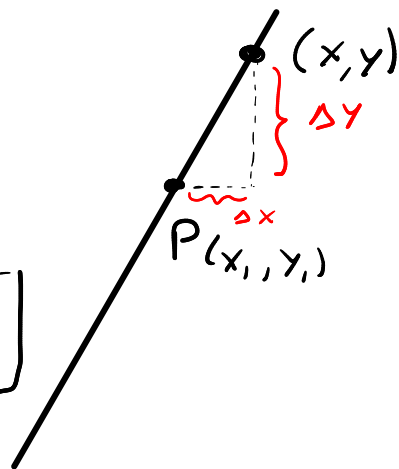
$$m_{\overline{AC}} \cdot m_{\overline{CB}} = -2, \text{ logo } \overline{AC} \perp \overline{CB} \quad (\theta = 90^\circ)$$

Equações da
Reta tendo um
Ponto (x_1, y_1) e
o coef. angular (m)

$$\frac{y - y_1}{x - x_1} = m$$

↓

$$y - y_1 = m(x - x_1)$$



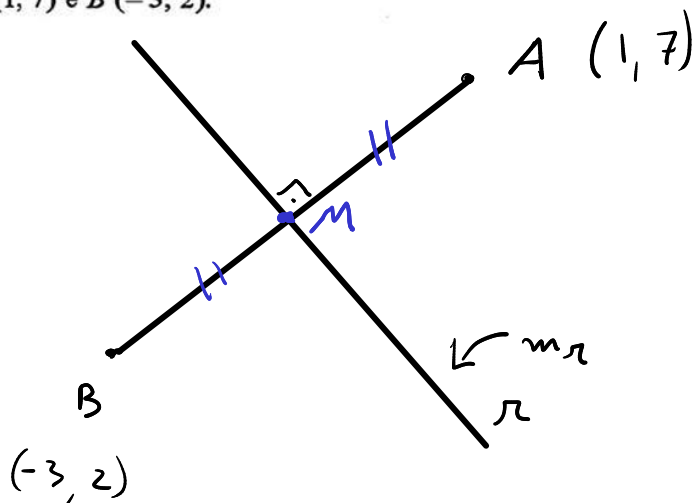
"Distribuído m ": $y - y_1 = mx - mx_1$

$$y = mx - \underbrace{mx_1 + y_1}_b \Leftrightarrow y = mx + \underbrace{b}$$

$$b = -mx_1 + y_1$$

Forma reduzida

Exemplo 4. Determinar a equação da bissetriz perpendicular do segmento de reta definido por $A(1, 7)$ e $B(-3, 2)$.



r é bissetriz perpendicular de AB

M é ponto médio de $\overline{AB}$

1º) Conf. angular de $\overleftrightarrow{AB} = m_{AB}$

$$m_{AB} = \frac{\Delta Y}{\Delta X} = \frac{7-2}{1-(-3)} = \frac{5}{4} \Rightarrow m_{AB} = \frac{5}{4}$$

$$r \perp \overleftrightarrow{AB} \Rightarrow m_r \cdot m_{AB} = -1$$

$$m_r \cdot \frac{5}{4} = -1 \Rightarrow m_r = -\frac{4}{5}$$

2º) Ponto médio:

$$M = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2} \right) = \left(-1, \frac{9}{2} \right)$$

$$\left. \begin{array}{l} \text{reta } r: m_r = -\frac{4}{5} \\ (-1, \frac{9}{2}) \in r \end{array} \right\} \Rightarrow \begin{array}{l} \text{Eq. da reta} \\ y - y_1 = m(x - x_1) \\ y - \frac{9}{2} = -\frac{4}{5}(x - (-1)) \end{array}$$

Eq. da reta bissetriz da AB :

$$y - \frac{9}{2} = -\frac{4}{5}(x + 1)$$

Exercícios

S: 1.3 | 1, 4, 8, 11, 14, 15, 25, 27, 31, 35, 36
37, 38, 39 e 40